MP0: Setting Up a Kernel Development Environment

CS 423: Operating System Design

Fall 2026

Important Dates

- **MP0 is released today.**
- **The due date is 9/14 11:59:59 PM CT.**
- **No late submission, even 1 second late.**

Goals

- You will configure your development environment for upcoming MPs.
- You will download, compile, and test your own kernel.
- The kernel source code will be a helpful reference.

What you need

- A Linux environment to compile your kernel and develop the MP.
- Another Linux environment to test your kernel/module.

CS 423 Infra

Qemu VM	Qemu VM	Qemu VM	......	Qemu VM
CS423 VM 01	CS423 VM 02	CS423 VM 03		CS423 VM xx

Engineering IT CS 423 VM Farm

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CS 423 VM

- For your development
- Provided by EngrIT
- You must use this VM to complete the MPs
- Ubuntu 24.04, No GUI

CS 423 Qemu

- **Another VM for testing your kernel/module**
- **Q script creates an overlayfs on your VM, modification not saved**
- **Crash won’t affect the host VM/your development, fast reboot**
- **… Yes, it is nested virtualization**

Connecting to your VM

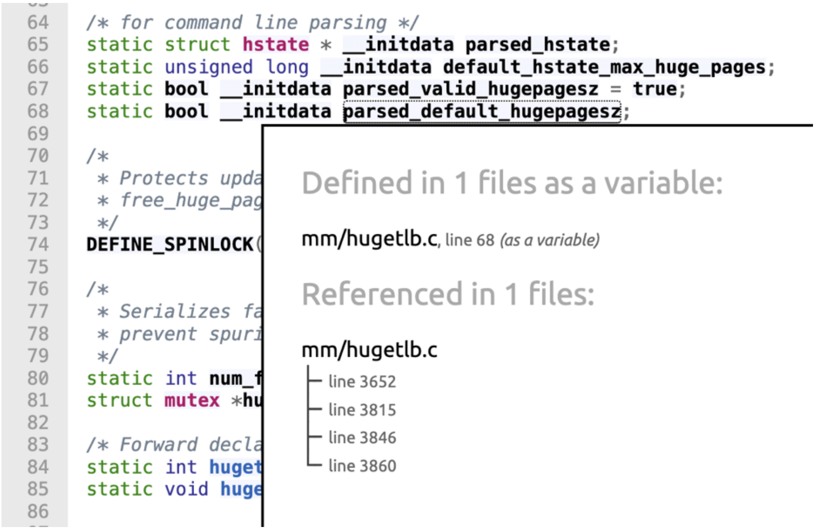
- Check your arrangement and power it ON first
- Use your favorite SSH client or tool like VSCode
- Install your public key (typing NetID password everytime?)
- We strongly recommend VSCode…

Demo

Navigating your code/kernel source

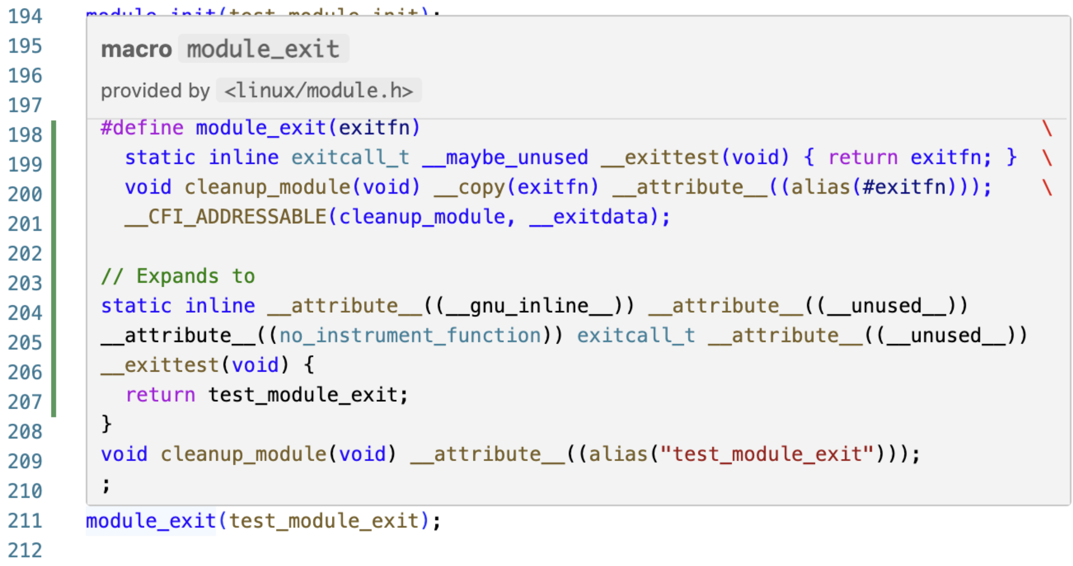
- **Bootlin Elixir (Web)**

  https://elixir.bootlin.com/linux/



- **VSCode + clangd (recommended)**

  Needs some work to make it functional



Submission

- **Submit to Google Form (posted on the MP0 doc).**
- `dmesg | grep 'Linux version’`
- **Requires Illinois login.**
- **We will use your last submission for grading.**

Q/A